

Diet and sex-hormone binding globulin, dysmenorrhea, and premenstrual symptoms.



OBJECTIVE: To test the hypothesis that a low-fat, vegetarian diet reduces dysmenorrhea and premenstrual symptoms by its effect on serum sex-hormone binding globulin concentration and estrogen activity. METHODS: In a crossover design, 33 women followed a low-fat, vegetarian diet for two menstrual cycles. For two additional cycles, they followed their customary diet while taking a supplement placebo pill. Dietary intake, serum sex-hormone binding globulin concentration, body weight, pain duration and intensity, and premenstrual symptoms were assessed during each study phase. RESULTS: Mean (+/- standard deviation [SD]) serum sex-hormone binding globulin concentration was higher during the diet phase (46.7 +/- 23.6 nmol/L) than during the supplement phase (39.3 +/- 19.8 nmol/L, $P < .001$). Mean (+/- SD) body weight was lower during the diet (66.1 +/- 11.3 kg) compared with the supplement phase (67.9 +/- 12.1 kg, $P < .001$). Mean dysmenorrhea duration fell significantly from baseline (3.9 +/- 1.7 days) to diet phase (2.7 +/- 1.9 days) compared with change from baseline to supplement phase (3.6 +/- 1.7 days, $P < .01$). Pain intensity fell significantly during the diet phase, compared with baseline, for the worst, second-worst, and third-worst days, and mean durations of premenstrual concentration, behavioral change, and water retention symptoms were reduced significantly, compared with the supplement phase. CONCLUSION: A low-fat vegetarian diet was associated with increased serum sex-hormone binding globulin concentration and reductions in body weight, dysmenorrhea duration and intensity, and premenstrual symptom duration. The symptom effects might be mediated by dietary influences on estrogen activity.